# Christopher M. **Ushay, Ph.D.**

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## Summary

Princeton-trained engineering PhD with deep fluency in materials science and soft matter physics paired with experience spanning early-stage startups, award-winning teaching, and cross-functional team leadership. Has driven complex projects from dense, ambiguous questions to validated outcomes and translated findings into narratives for both scientific and non-technical audiences. Seeking roles spanning deep tech, analytics, and project management.

## Education

**Princeton University** *Princeton, NJ*

**Ph.D., Chemical & Biological Engineering**

- Dissertation - "Leveraging Relationships Between Confined Flows and Deformable Media," under advisor Prof. Pierre-Thomas Brun

**William E. Macaulay Honors College, City University of New York** *New York, NY*

**Bachelor of Engineering, Chemical Engineering, Mathematics Minor**

- Valedictorian, Grove School of Engineering at City College of New York | Summa Cum Laude

## Work Experience

**Independent Contractor, Technical Instruction & AI Training** *Remote*

**Self-Employed** *January 2024 - Present*

- Engaged as a domain expert in applied physics, materials science, and engineering to develop and evaluate technical AI training content
- Produced precise, well-reasoned prompts and responses requiring both scientific fluency and clear written communication

**Doctoral Researcher** *Princeton, NJ*

**Lab of Pierre-Thomas Brun, Department of Chemical & Biological Engineering, Princeton University** *January 2019 - May 2024*

- Owned multi-year research programs end-to-end by scoping open problems in physics, setting roadmaps, coordinating collaborators across multiple institutions, and delivering validated findings against hard deadlines for publication and conferences
- Transformed technically dense, ambiguous research into a dissertation exploring open questions in mechanics and materials science by distilling research findings into short- and long-form narratives, including multiple peer-reviewed articles in high-impact journals and a cover page feature
- Wrote and automated Python and Mathematica pipelines to convert raw experimental measurements into numerical data, analyze large datasets against theoretical predictions, and visualize relationships to glean defensible, publication-grade insights
- Developed 2D and 3D multiphysics models of soft material and fluid-structure systems, using simulation to validate experimental observations and explore design space
- Communicated research findings to global scientific audiences through nearly 10 presentations at major physics conferences and multi-lab collaborations, supported by original figures and multimedia produced in Adobe CC, AutoCAD, and Keynote
- Contributed to scientific art pieces, movies, and sculptures inspired by the lab's research

**Assistant in Instruction** *Princeton, NJ*

**School of Engineering & Applied Science, Princeton University** *January 2020 - December 2022*

- Recognized with Princeton University's annual Excellence in Teaching Award on student nomination for the instruction of Undergraduate Differential Equations (Spring '20, '21) and Graduate Soft Matter Mechanics (Fall '22), reflecting consistent investment towards accessibility of complex material
- Taught classes of 70+ undergraduate students introductory to advanced topics in math and physics, spanning classical theory and modern computational tools, by leading weekly lectures, office hours, and ongoing availability as a dedicated resource

**Head Manager** *Princeton, NJ*

**Debasement Bar, Student-Run Hospitality Venture, Princeton University** *July 2021 - December 2023*

- Managed full P&L operations of a hospitality business, overseeing a 20+ person team, regulatory compliance, and financial reporting with revenue on the order of tens of thousands
- Maintained facilities (equipment, consumables) and developed data-driven longevity plans with co-managers to ensure continued improvement beyond tenures, including programs to enhance experiences such as the implementation of ride-share vouchers for patrons
- Programmed nightly events for hundreds of graduate students and affinity groups in order to provide gathering space and foster community

**Machine Learning Research Assistant** *New York, NY*

**Element, Inc.** *October 2017 - August 2018*

- Operated inside company's $12M Series A growth phase with cross-functional exposure across Operations and Engineering, including hiring, onboarding, and company culture development, during a period of rapid headcount and process change
- Built and ran data pipelines to process and filter biometric data to support an ML identification platform targeting underserved markets in developing economies

## Skills & Interests

- **Software & Data Science** - Python (inc. NumPy, pandas, Matplotlib, PyTorch); Mathematica; SQL; COMSOL; MATLAB; C; Julia
- **Fabrication & Analysis** - Laser cutting; CNC milling; 3D printing; digital photography; high-speed imaging; microscopy
- **Design & Visualization** - Adobe CC (Illustrator, Photoshop); Keynote/PowerPoint; AutoCAD; Fiji

- **Languages** - English (fluent); French (CEFR C1); Spanish (CEFR C1); German (CEFR A2); Japanese (elementary)

## Publications

- **Ushay, C.**, Jambon-Puillet, E., Brun, P.-T. (2023) "Interfacial flows through arrays of elastic fibers" Physical Review Fluids, DOI: 10.1103/PhysRevFluids.8.044001
- Badaoui, M., Kresge, G., **Ushay, C.**, Marthelot, J., Brun, P.-T. (2022) "Formation of Pixelated Elastic Films via Capillary Suction of Curable Elastomers in Templated Hele–Shaw Cells" Advanced Materials, DOI: 10.1002/adma.202109682
- Travlou, N.A., **Ushay, C.**, Seredych, M., Rodriguez-Castellon, E., Bandosz, T.J. (2016) "Nitrogen-Doped Activated Carbon-Based Ammonia Sensors: Effect of Specific Surface Functional Groups on Carbon Electronic Properties" ACS Sensors, DOI: 10.1021/acssensors.6b00093

## Leadership & Honors

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| 2026 | **Board Member**, Princeton University Graduate Alumni Association | *Remote* |
| 2022 | **Cover Page Feature**, Advanced Materials, Volume 34, Issue 27 | *Print & Online* |
| 2021 | **Excellence in Teaching Award**, Princeton University | *Princeton, NJ* |
| 2017 | **Valedictorian**, Grove School of Engineering, City College of New York | *New York, NY* |
| 2017 | **Alumni Medal for Highest Departmental GPA**, Grove School of Engineering, City College of New York | *New York, NY* |

## Presentations, Exhibitions, & Press

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| 2024-25 | **Rope Piece (Deluge); Deluge 2; Deluge 3**, Traveling Gallery of Fluid Motion: Spiraling Upwards | *Salt Lake City, UT*<br>***Sculpture*** |
| 2023 | **"Elastocapillary painting: deposition of fluid trapped within fibre arrays"**, American Physical Society March Meeting 2023 | *Las Vegas, NV*<br>***Presentation*** |
| 2022 | **"Hairy fluid mechanics: Dynamic elastocapillarity in deformable beam arrays"**, 75th Annual Meeting of the Division of Fluid Dynamics | *Indianapolis, IN*<br>***Presentation*** |
| 2022 | **"Fluid mediated assembly of pixelated materials"**, 75th Annual Meeting of the Division of Fluid Dynamics | *Indianapolis, IN*<br>***Presentation*** |
| 2022 | **"Hairy fluid mechanics: interfacial flows through networks of flexible fibres"**, 19th U.S. National Congress on Theoretical and Applied Mechanics | *Austin, TX*<br>***Presentation*** |
| 2022 | **"Easy as an inkjet, a new soft printing technique has opened the way for pixelated elastics"**, Princeton University Communications | *Online*<br>***Article*** |
| 2021 | **"Hairy fluid mechanics: interfacial flows in networks of flexible fibres"**, 74th Annual Meeting of the Division of Fluid Dynamics | *Phoenix, AZ*<br>***Presentation*** |
| 2020 | **"Fluid Capture Patterns in Hairy Surfaces"**, 73rd Annual Meeting of the Division of Fluid Dynamics | *Online*<br>***Presentation*** |
| 2020 | **"Multiphase Flow Through Hairy Channels"**, American Physical Society March Meeting 2020 | *Online*<br>***Presentation*** |